

Week 10: Lab

INFT 1103: Operating System Security

Durham College

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November 26, 2025

Week 10 Lab – Creating and Adjusting AppArmor Profiles

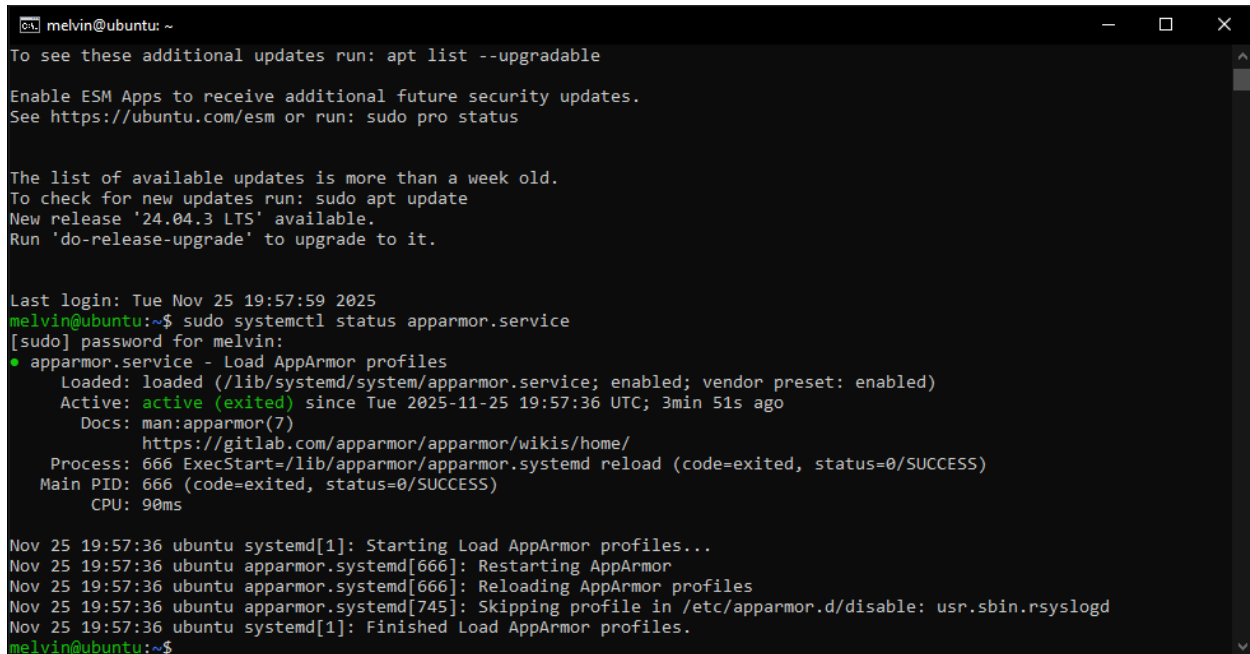
In this lab, you will be creating and adjusting AppArmor profile settings and testing the difference between enforce and complain modes.

*Please remember to use the snipping tool to take a screenshot of the relevant information to each question only. **Illegible screenshots will not be graded.***

Include this outline in your submission. Any submissions not including the content of this outline, or those that include any fabricated content, will receive a grade of 0.

1. Check that Apparmor is currently installed and running on your machine.

Sudo systemctl status apparmor.service

A terminal window titled 'melvin@ubuntu: ~' showing the output of 'sudo systemctl status apparmor.service'. The output includes system update notifications, the last login time, and detailed status for the apparmor.service, which is active (exited). It also shows a series of logs from systemd indicating the successful loading and reloading of AppArmor profiles.

```
melvin@ubuntu: ~  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
New release '24.04.3 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Last login: Tue Nov 25 19:57:59 2025  
melvin@ubuntu:~$ sudo systemctl status apparmor.service  
[sudo] password for melvin:  
• apparmor.service - Load AppArmor profiles  
   Loaded: loaded (/lib/systemd/system/apparmor.service; enabled; vendor preset: enabled)  
   Active: active (exited) since Tue 2025-11-25 19:57:36 UTC; 3min 51s ago  
     Docs: man:apparmor(7)  
           https://gitlab.com/apparmor/apparmor/wikis/home/  
   Process: 666 ExecStart=/lib/apparmor/apparmor.systemd reload (code=exited, status=0/SUCCESS)  
  Main PID: 666 (code=exited, status=0/SUCCESS)  
    CPU: 90ms  
  
Nov 25 19:57:36 ubuntu systemd[1]: Starting Load AppArmor profiles...  
Nov 25 19:57:36 ubuntu apparmor.systemd[666]: Restarting AppArmor  
Nov 25 19:57:36 ubuntu apparmor.systemd[666]: Reloading AppArmor profiles  
Nov 25 19:57:36 ubuntu apparmor.systemd[745]: Skipping profile in /etc/apparmor.d/disable: usr.sbin.rsyslogd  
Nov 25 19:57:36 ubuntu systemd[1]: Finished Load AppArmor profiles.  
melvin@ubuntu:~$
```

Sudo aa-status

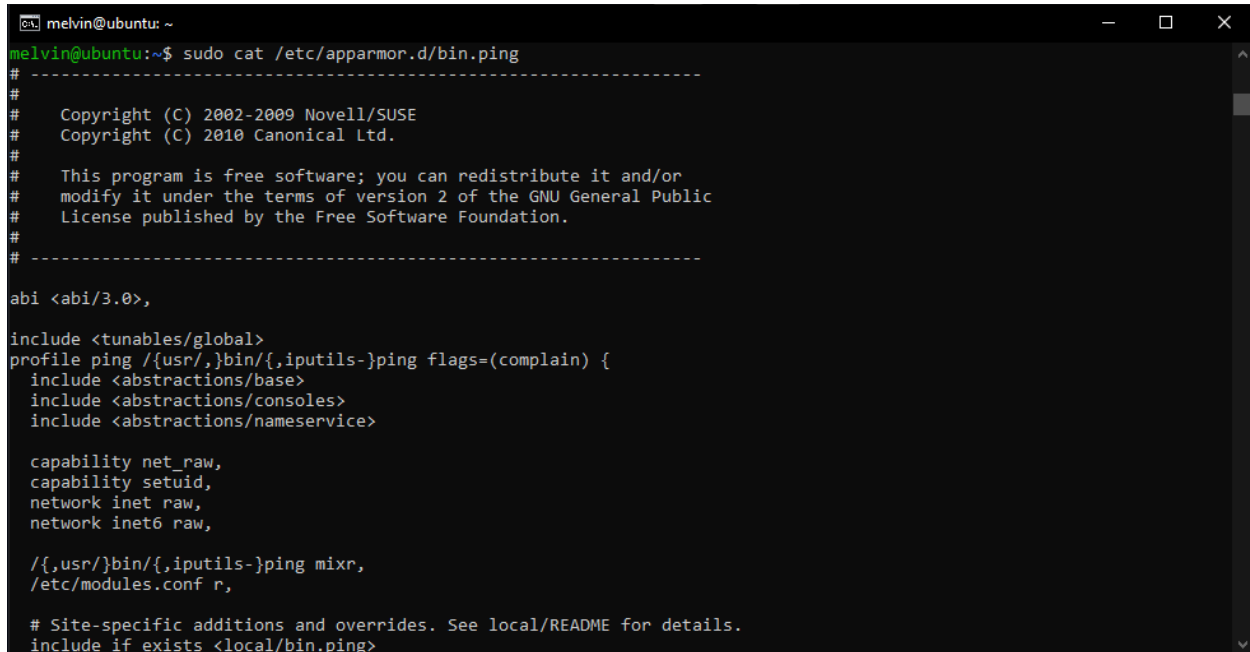
```
melvin@ubuntu: ~  
melvin@ubuntu:~$ sudo aa-status  
apparmor module is loaded.  
58 profiles are loaded.  
41 profiles are in enforce mode.  
/snap/snapd/21759/usr/lib/snapd/snap-confine  
/snap/snapd/21759/usr/lib/snapd/snap-confine//mount-namespace-capture-helper  
/usr/bin/freshclam  
/usr/bin/man  
/usr/lib/NetworkManager/nm-dhcp-client.action  
/usr/lib/NetworkManager/nm-dhcp-helper  
/usr/lib/connman/scripts/dhclient-script  
/usr/lib/snapd/snap-confine  
/usr/lib/snapd/snap-confine//mount-namespace-capture-helper  
/usr/sbin/clamd  
/{usr}/sbin/dhclient  
lsb_release  
man_filter  
man_groff  
nvidia_modprobe  
nvidia_modprobe//kmod  
snap-update-ns.lxd  
snap.lxd.activate  
snap.lxd.benchmark  
snap.lxd.buginfo  
snap.lxd.check-kernel  
snap.lxd.daemon  
snap.lxd.hook.configure  
snap.lxd.hook.install  
snap.lxd.hook.remove  
snap.lxd.lxc  
snap.lxd.lxc-to-lxd  
snap.lxd.lxd  
snap.lxd.migrate  
snap.lxd.user-daemon  
tcpdump  
ubuntu_pro_apt_news  
ubuntu_pro_esm_cache  
ubuntu_pro_esm_cache//apt_methods  
ubuntu_pro_esm_cache//apt_methods_gpgv  
ubuntu_pro_esm_cache//cloud_id  
ubuntu_pro_esm_cache//dpkg  
ubuntu_pro_esm_cache//ps  
ubuntu_pro_esm_cache//ubuntu_distro_info  
ubuntu_pro_esm_cache_systemctl  
ubuntu_pro_esm_cache_systemd_detect_virt  
17 profiles are in complain mode.  
avahi-daemon  
dnsmasq  
dnsmasq//libvirt_leaseshelper  
identd  
klogd  
mdnsd  
nmbd  
nscd  
php-fpm  
ping  
samba-bgqd  
smbd  
smbldap-useradd  
smbldap-useradd///etc/init.d/nscd  
syslog-ng  
syslogd  
traceroute  
0 profiles are in kill mode.  
0 profiles are in unconfined mode.  
1 processes have profiles defined.  
1 processes are in enforce mode.  
/usr/bin/freshclam (927)  
0 processes are in complain mode.  
0 processes are unconfined but have a profile defined.  
0 processes are in mixed mode.  
0 processes are in kill mode.  
melvin@ubuntu:~$
```

2. View the apparmor profile for the ping command:

Sudo cat /etc/apparmor.d/bin.ping

Note the capability “net_raw” and “setuid” – these capabilities allow the ping process to work by sending and receiving packets

Take a screenshot of these settings and paste it here



```
melvin@ubuntu: ~  
melvin@ubuntu:~$ sudo cat /etc/apparmor.d/bin.ping  
# -----  
#  
# Copyright (C) 2002-2009 Novell/SUSE  
# Copyright (C) 2010 Canonical Ltd.  
#  
# This program is free software; you can redistribute it and/or  
# modify it under the terms of version 2 of the GNU General Public  
# License published by the Free Software Foundation.  
#  
# -----  
  
abi <abi/3.0>,  
  
include <tunables/global>  
profile ping /{usr/,}bin/{,iputils-}ping flags=(complain) {  
  include <abstractions/base>  
  include <abstractions/consoles>  
  include <abstractions/nameservice>  
  
  capability net_raw,  
  capability setuid,  
  network inet raw,  
  network inet6 raw,  
  
  /{,usr/}bin/{,iputils-}ping mixr,  
  /etc/modules.conf r,  
  
  # Site-specific additions and overrides. See local/README for details.  
  include if exists <local/bin.ping>  
}
```

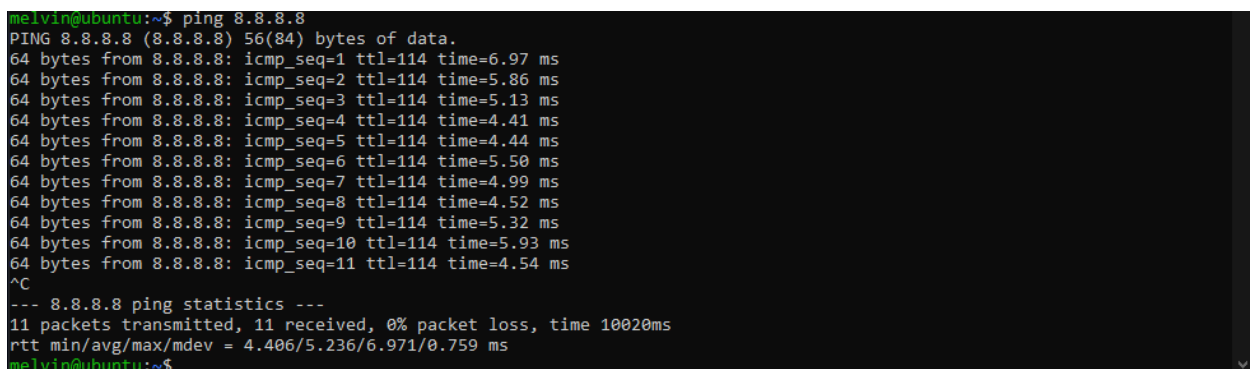
3. Ping an external address by running the command

Ping 8.8.8.8

Make sure that you receive a response.

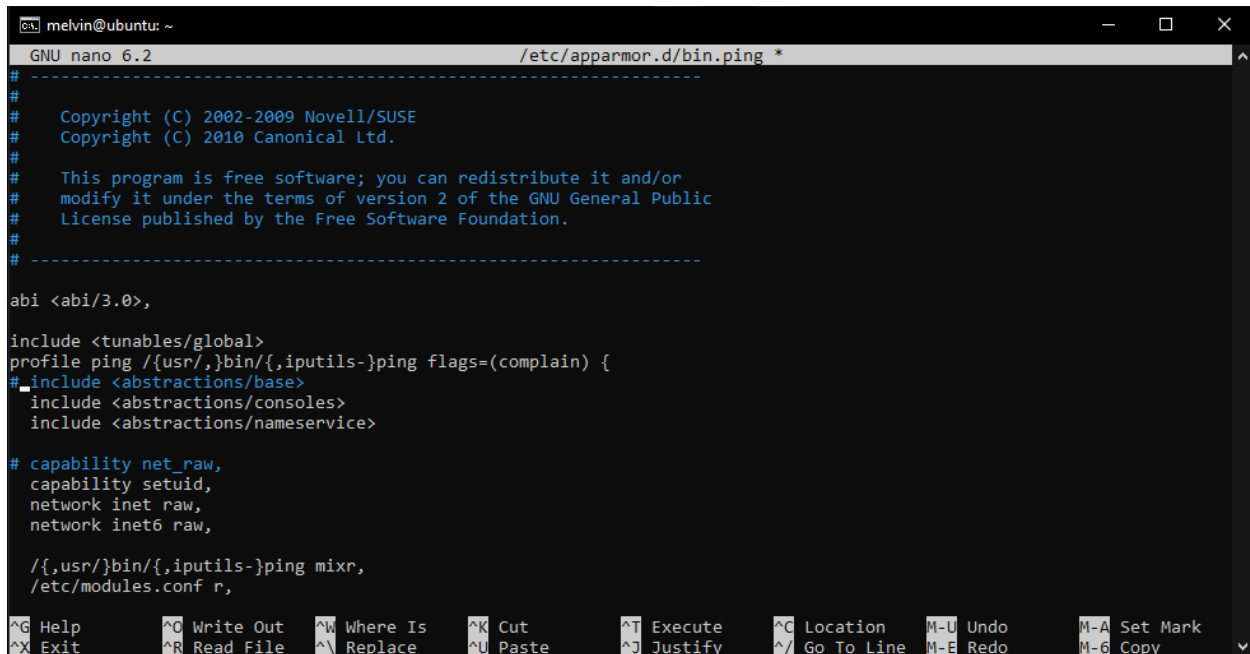
Allow the ping to run for at least 4 packets, then end the command using ctrl+c

Take a screenshot of your results and paste it here



```
melvin@ubuntu:~$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8): 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=6.97 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=5.86 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=5.13 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=4.41 ms  
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=4.44 ms  
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=5.50 ms  
64 bytes from 8.8.8.8: icmp_seq=7 ttl=114 time=4.99 ms  
64 bytes from 8.8.8.8: icmp_seq=8 ttl=114 time=4.52 ms  
64 bytes from 8.8.8.8: icmp_seq=9 ttl=114 time=5.32 ms  
64 bytes from 8.8.8.8: icmp_seq=10 ttl=114 time=5.93 ms  
64 bytes from 8.8.8.8: icmp_seq=11 ttl=114 time=4.54 ms  
^C  
--- 8.8.8.8 ping statistics ---  
11 packets transmitted, 11 received, 0% packet loss, time 10020ms  
rtt min/avg/max/mdev = 4.406/5.236/6.971/0.759 ms  
melvin@ubuntu:~$
```

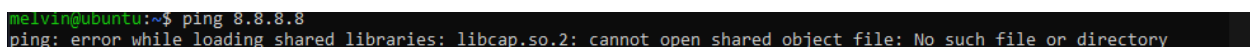
4. Open the *ping* apparmor profile in nano:
Sudo nano /etc/apparmor.d/bin.ping
5. Make the following adjustments to the profile:
 - a. Comment out the line: *include <abstractions/base>*
 - b. Comment out the line: *capability net_raw**Take a screenshot of your adjustments and paste it here*



```
melvin@ubuntu: ~  
GNU nano 6.2 /etc/apparmor.d/bin.ping *  
#-----  
# Copyright (C) 2002-2009 Novell/SUSE  
# Copyright (C) 2010 Canonical Ltd.  
#  
# This program is free software; you can redistribute it and/or  
# modify it under the terms of version 2 of the GNU General Public  
# License published by the Free Software Foundation.  
#-----  
abi <abi/3.0>,  
  
include <tunables/global>  
profile ping /{usr/,}bin/{,iputils-}ping flags=(complain) {  
#_include <abstractions/base>  
  include <abstractions/consoles>  
  include <abstractions/nameservice>  
  
# capability net_raw,  
  capability setuid,  
  network inet raw,  
  network inet6 raw,  
  
  /{,usr/}bin/{,iputils-}ping mixr,  
  /etc/modules.conf r,  
}
```

6. Close the ping profile and save the changes by pressing Y.
7. Place the profile into enforce mode:
sudo aa-enforce /etc/apparmor.d/bin.ping
8. Restart the AppArmor ping profile:
sudo apparmor_parser -r /etc/apparmor.d/bin.ping
9. Ping an external address:
ping 8.8.8.8

Take a screenshot of your results and paste them here



```
melvin@ubuntu:~$ ping 8.8.8.8  
ping: error while loading shared libraries: libcap.so.2: cannot open shared object file: No such file or directory
```

10. Place the profile into complain mode:
sudo aa-complain /etc/apparmor.d/bin.ping

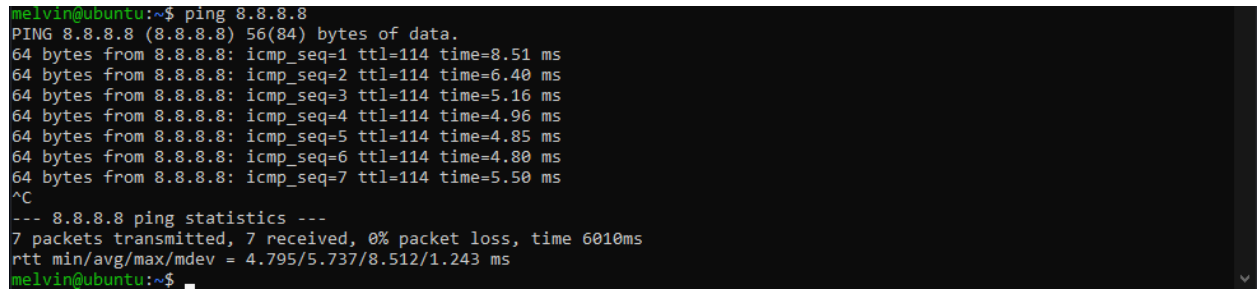
11. Restart the AppArmor ping profile:

```
sudo apparmor_parser -r /etc/apparmor.d/bin.ping
```

12. Ping an external address:

```
ping 8.8.8.8
```

Take a screenshot of your results and paste them here



```
melvin@ubuntu:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=8.51 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=6.40 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=5.16 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=4.96 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=4.85 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=4.80 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=114 time=5.50 ms
^C
--- 8.8.8.8 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6010ms
rtt min/avg/max/mdev = 4.795/5.737/8.512/1.243 ms
melvin@ubuntu:~$
```

13. Reopen your Ping Apparmor profile:

```
Sudo nano /etc/apparmor.d/bin.ping
```

14. Revert the changes made in step 6.

15. Exit the profile and save the changes using Y.

16. Restart the AppArmor profile:

```
sudo apparmor_parser -r /etc/apparmor.d/bin.ping
```